

Rossmann Store Sales Forecasting

Project Summary Report — Time Series Sales Prediction

Objective

Forecast daily store sales using historical sales patterns, promotions, and store attributes, to help retailers plan inventory, staffing, and promotional strategy more effectively.

Dataset

Source	Rossmann Store Sales (Kaggle)
Records	1,017,209 daily sales records
Stores	1,115 unique stores
Time Period	Jan 2013 – Jul 2015

Key Insights from Exploratory Data Analysis

- Promotions increase average daily sales by **38.8%** (■5,929 → ■8,228 average).
- Weekly pattern: sales peak on **Monday and Sunday** (~■8,200 avg); lowest on **Saturday** (~■5,875 avg).
- Store Type B** outperforms all other store types, averaging ■10,231 vs. ~■6,900 for types A, C, and D.
- 98%+ of days are non-holidays; holiday-specific effects are minor relative to promotions and weekly patterns.

Feature Engineering

- Date-based features: year, month, day, week of year, weekend flag
- Competition features: months since competitor opened, competitor distance
- Lag features: previous day's sales, previous week's sales
- Rolling average: 7-day rolling mean of sales
- One-hot encoding: store type, assortment, holiday type

Model & Results

Algorithm: **Random Forest Regressor** | Split: time-based 80/20 (no random shuffling, preserving chronology)

Metric	Value	Interpretation
R ² Score	0.874	Model explains 87.4% of the variance in daily sales
MAE	741.92	Average prediction error of ~742 units (~13% of avg. sales)

RMSE	1,087.66	Penalizes larger errors; within acceptable range
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Top Predictive Features

Feature	Importance
Previous day's sales (Sales_Lag1)	62.3%
Day of week	10.3%
7-day rolling average	9.7%
Promotion active	6.2%
Day of month	2.7%

Takeaway: Recent sales history (yesterday's sales and the 7-day trend) accounts for over 70% of the model's predictive power — short-term momentum matters far more than any single static store attribute.

Business Recommendations

- Shift some promotional activity to **Saturdays** to smooth out the weekly demand dip.
- Ensure adequate **stock and staffing ahead of Mondays and Sundays**, the highest-traffic days.
- Study operational practices at **Store Type B** locations — they may offer a replicable template for other store types.

Tech Stack

Python · Pandas · NumPy · Matplotlib · Seaborn · Scikit-learn (Random Forest)

GitHub Repository: github.com/asthapandey-ext/Rossman_Sales_Forecasting